

A FIVE-MINUTE QUANTUM STORY

Can something pass through a wall?

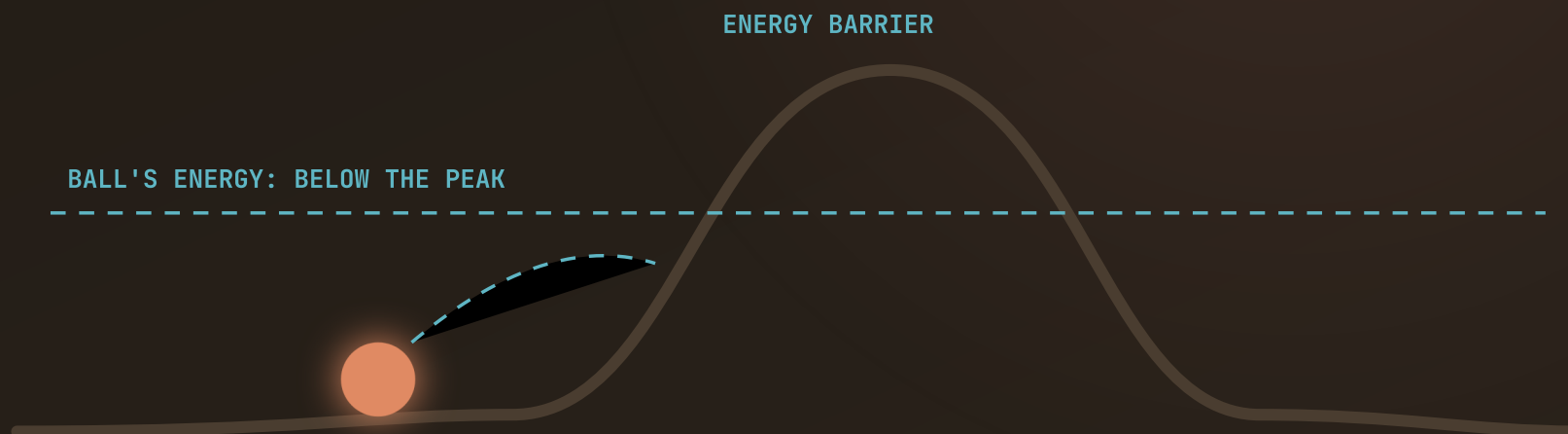
In everyday life, never. In the quantum world, **sometimes**.

ENERGY BARRIER



FIRST: THE FAMILIAR WORLD

A ball needs enough energy to cross a hill.

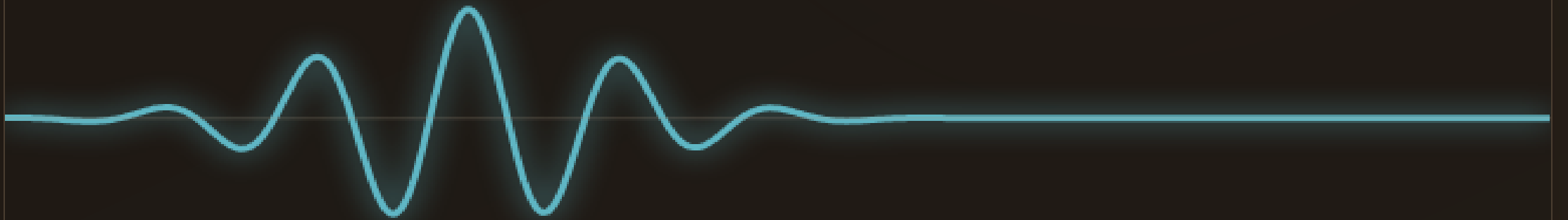


NOW SHRINK EVERYTHING

A particle is not just a tiny ball.

An electron is described by a [wave of possibilities](#), showing where it might be found.

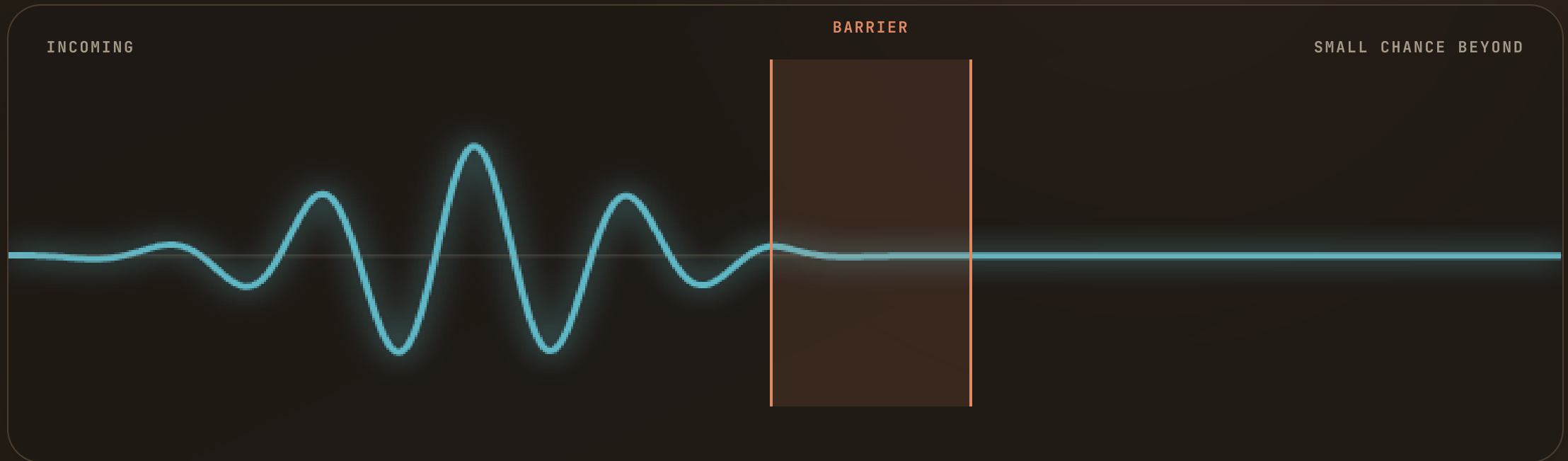
PROBABILITY WAVE



THE KEY IDEA

The wave does not stop instantly.

Inside the barrier it fades, but may not reach zero.



NO SECRET TUNNEL

Measure many identical particles...

MOST

bounce back

The particle is detected before the barrier.

A FEW

appear beyond

They are detected on the other side.

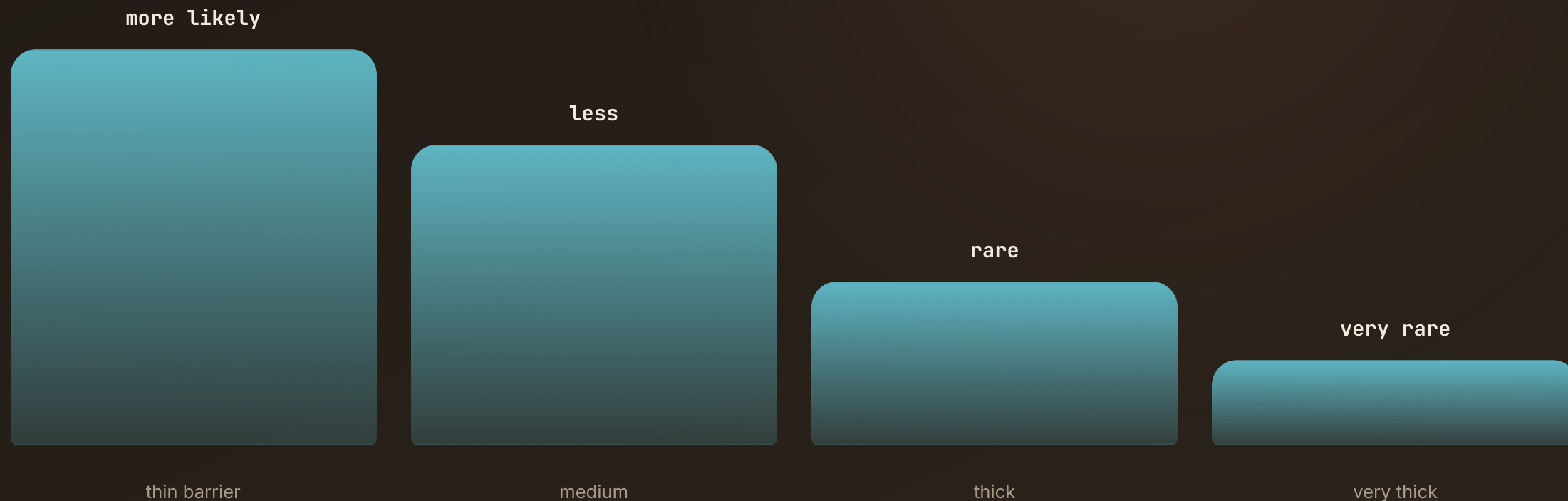
NEVER

half a particle

Each measurement finds one whole particle.

WHAT CHANGES THE CHANCE?

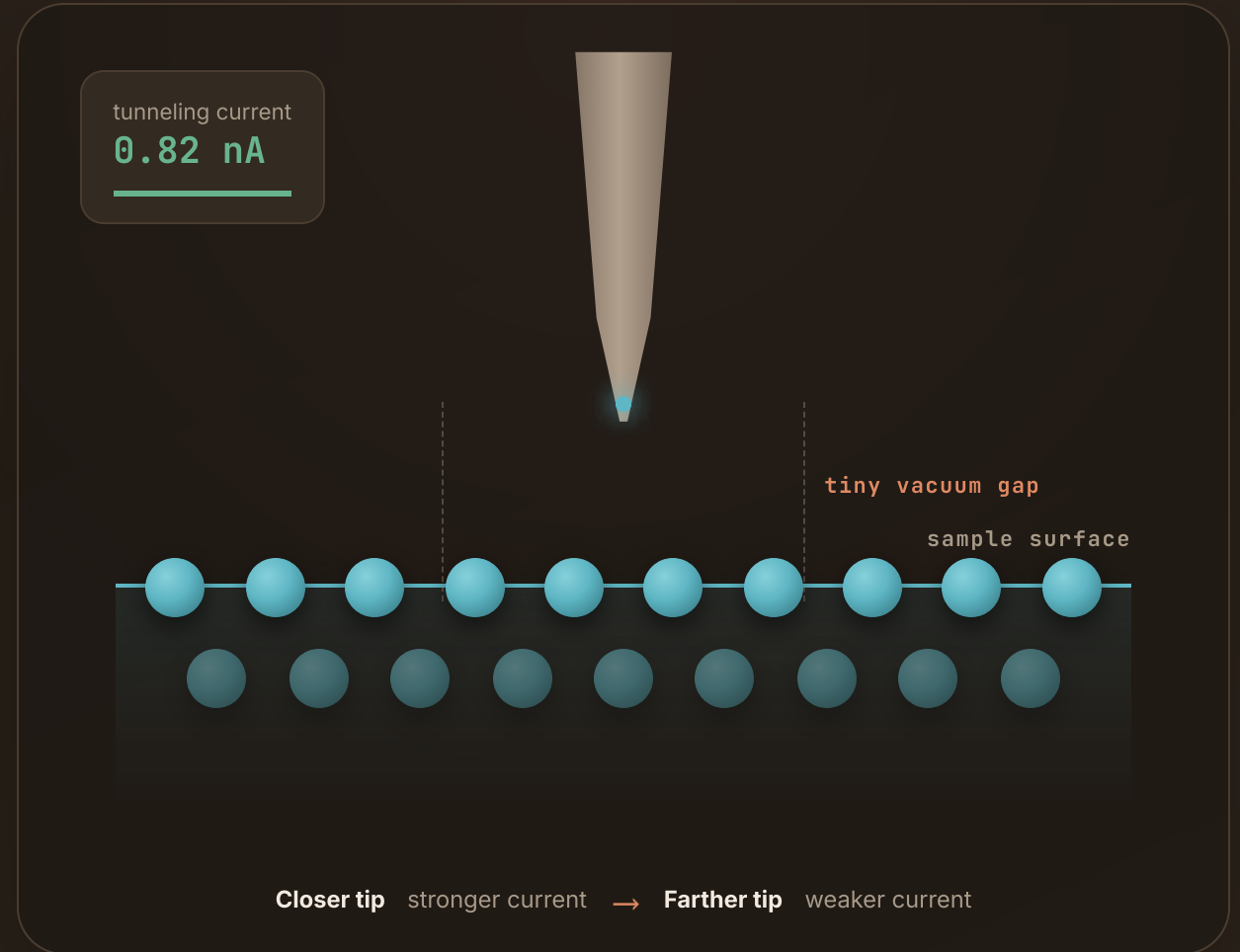
Thinner and lower barriers are easier to tunnel through.



A MICROSCOPE BUILT ON TUNNELING

A tiny current maps individual atoms.

In a scanning tunneling microscope, electrons cross a microscopic gap. The current changes dramatically with distance, turning the surface into an image.



NOT JUST A THOUGHT EXPERIMENT

Tunneling quietly shapes our world.



Stars

Helps nuclei get close enough for fusion.



Electronics

Enables devices and creates leakage in tiny chips.



Microscopy

Lets us image surfaces atom by atom.

● A particle can cross
a barrier it cannot
classically cross.

Because quantum objects behave like waves of probability.

Impossible classically. Possible quantum mechanically.